From Reconstruction to Anticipation: Self-Supervised Prediction of Future Tissue Geometry in Monocular Laparoscopic Video

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***Abstract*—Four-dimensional reconstruction of endoscopic scenes has advanced rapidly, and recent Gaussian-splatting methods recover deforming tissue at high fidelity in real time. Every such system, however, is *retrospective*: it describes deformation that has already been observed and offers no estimate of deformation that has not. Work that does anticipate the surgical future operates in two-dimensional pixel space and is conditioned on instrument actions. We present, to our knowledge, the first pipeline that forecasts future tissue *geometry* from monocular laparoscopic video without any clinical, action or instrument annotation. Our system reconstructs the scene as a deformable Gaussian field, extracts a time-consistent trajectory of 512 tracked tissue points, converts that trajectory into a self-supervised dataset of 5,910 windows, and trains a conditional denoising diffusion model conditioned jointly on observed geometry and on the requested prediction horizon. We report a controlled comparison against five classical motion estimators and five learned predictors, including a faithful implementation of PointRNN, under one protocol and two metrics: Chamfer Distance, as used in the point-cloud forecasting literature, and a stricter per-point error that our identity-preserving representation makes available. Reconstruction reaches 30.10 dB PSNR from monocular input alone, within the range reported by stereo-supervised methods on comparable sequences. On prediction we report a negative result, analysed in detail: the model recovers the *direction* of tissue motion but not its magnitude, which at a displacement scale of 4×10^{-3} normalised units is sufficient to forfeit the accuracy comparison. The sampled futures are nevertheless well calibrated ($r = 0.69$ between predicted spread and realised error), giving a usable indication of when the system should not be trusted.**

***Index Terms*—surgical vision, 4D reconstruction, Gaussian splatting, diffusion models, deformation forecasting, self-supervised learning, uncertainty calibration**

I. Introduction

Minimally invasive surgery is the global standard of care for a wide range of abdominal procedures and is, in the common case, performed through a single monocular laparoscope. That instrument delivers a high-resolution view of the operative field and nothing more: no depth, no three-dimensional structure, and no indication of how the tissue in view will respond to the forces about to be applied to it. Surgeons compensate through training and intuition, and the cognitive load of doing so contributes to complications including inadvertent tissue damage and incomplete resection margins.

Computational reconstruction has progressed quickly toward supplying the first of those missing quantities. Neural radiance fields were adapted to deformable endoscopic tissue [1], accelerated by factored spatio-temporal representations [2], and then largely superseded by 3D Gaussian Splatting [3], whose surgical variants [4], [5] reconstruct deforming tissue at real-time rates and high photometric fidelity.

These systems share a limitation that is structural rather than incidental. All of them are retrospective. They answer what the tissue *did*, given video of it doing so. None answers what the tissue is *about to do*. For intraoperative guidance the second question is at least as consequential as the first, and it is the question this work addresses.

A. Why this is hard

Three properties distinguish the problem from superficially similar tasks. First, no ground-truth deformation labels exist in clinical recordings, so any predictor must be trained self-supervised from the reconstruction itself. Second, the quantity to be predicted is extremely small: inter-frame tissue displacement in our data averages 4×10^{-3} normalised units, one to two pixels when projected, against the unit-scale noise of a generative process. Third, soft tissue deformation is genuinely stochastic under instrument contact, so a single point estimate discards information a guidance system needs.

B. Contributions

1) **A complete pipeline from monocular video to forecast geometry**, spanning reconstruction and prediction, trained without clinical, action or instrument annotation (Section III).
2) **A conditional diffusion formulation over geometric displacement**, conditioned jointly on observed geometry and the prediction horizon, with a residual target that bounds the predictor's failure mode from below by a classical estimator (Section III-E).
3) **A controlled comparative study** against five classical estimators and five learned predictors — including PointRNN [6], the closest published method — under a single protocol, reported with both a correspondence-free and a correspondence-aware metric (Sections IV–V).

4) **A calibrated uncertainty estimate** and a failure analysis localising the residual error to displacement magnitude rather than direction, which is a prescriptive rather than merely descriptive negative result (Sections V-F–VI).

II. Related Work and Positioning

A. Reconstruction of deformable surgical scenes

EndoNeRF [1] conditioned a radiance field on time to model tissue deformation from stereo input, requiring hours of optimisation per scene. LerPlane [2] factorised the 4D volume into orthogonal feature planes, reducing this to minutes. EndoGaussian [4] replaced the volumetric representation with Gaussian primitives and holistic initialisation, reaching real-time rendering, and Deform3DGS [5] introduced per-Gaussian flexible deformation modelling, reporting the strongest published quality on both the EndoNeRF and StereoMIS benchmarks. **We take this family as our first base line of work**: our reconstruction stage follows it, and Section V-A positions our monocular results against their stereo-supervised ones.

B. Anticipation in surgical video

A separate line predicts the surgical future directly in image space. Gao *et al.* [7] forecast future frames of robotic surgical video with a ternary-prior variational autoencoder. SurgVista [8] extends this to long-horizon, action-conditioned world modelling with explicit regularisation for plausible tissue dynamics. Both predict *pixels*, and both condition on instrument actions. Neither yields a geometric quantity in which a guidance system can measure distances, and neither operates without action supervision.

C. Point cloud sequence forecasting

The closest formulation of our task outside surgery is point cloud prediction. PointRNN [6] introduced a point-based spatiotemporally-local correlation that aggregates the previous frame's hidden states over spatial neighbours together with the displacement to each, substituting it for the matrix products of a recurrent cell. MoNet [9] added explicit motion features; PCPNet [10] replaced recurrence with a transformer; and Mersch *et al.* [11] applied 3D spatio-temporal convolutions to range images. All target outdoor LiDAR, where inter-frame motion is metres rather than millimetres, scenes are predominantly rigid, and point identity is *not* preserved between scans — which is why correspondence must be inferred geometrically and why Chamfer Distance [12] is the standard score. We implement PointRNN as our principal learned baseline.

D. Diffusion models for 3D and for forecasting

Denoising diffusion [13], with a cosine schedule [14] and deterministic sampling [15], is now a standard generative framework. Luo and Hu [16] first applied it to 3D point clouds, treating points as particles diffusing from a shape distribution to noise; **we take this as our second base line of work**, and our denoiser is a conditional, temporally-grounded instance of it. For forecasting specifically, diffusion has been applied to pedestrian and vehicle trajectories [17], [18], where its principal advantage is producing a distribution over futures rather than a point estimate. In the medical domain, Xie *et al.* [19] condition a point cloud diffusion model on a single X-ray projection to recover liver surface deformation — but they estimate the *current* deformed state from an observation of it, not a future state from past states. We adopt the v-parameterisation [20] and Min-SNR loss weighting [21], both of which prove necessary at our displacement scale.

E. The gap

Table I states the gap precisely rather than broadly. Surgical reconstruction is retrospective. Surgical anticipation predicts pixels and requires action labels. Point cloud forecasting targets rigid outdoor scenes without correspondence or uncertainty. Medical point cloud diffusion estimates present deformation rather than future deformation. *Forecasting future 3D tissue geometry from monocular surgical video, without annotation, with a calibrated uncertainty estimate*, occupies none of these cells, and is what this work contributes.

III. System

The pipeline comprises six stages. Stages 1–3 reconstruct; stages 4–6 predict.

A. Stage 1: Preprocessing and monocular depth

Frames are extracted at fixed stride, normalised to 640×360 and filtered by a sharpness criterion. Because no stereo pair or depth sensor is available, geometry is initialised from monocular depth priors [22]. Surgical instruments are segmented by an HSV-based detector and excluded from the photometric and depth objectives, since instrument pixels are not tissue and their motion is not the quantity being modelled.

B. Stage 2: Deformable 4D Gaussian reconstruction

The scene is represented by anisotropic 3D Gaussians [3] with degree-3 spherical-harmonic appearance. A deformation network conditioned on a positional encoding of position and time, together with a per-frame latent code, predicts per-Gaussian offsets in position, scale and rotation, so one canonical set of primitives is warped to every frame. Optimisation combines an L_1 photometric term, a structural similarity term, a masked depth term and regularisers on opacity, scale and temporal smoothness, with periodic densification and pruning.

C. Stage 3: Time-consistent trajectory extraction

Because the deformation network warps one canonical set of primitives, every Gaussian retains its identity across all frames. Evaluating the deformation at each timestep yields a trajectory tensor $\mathcal{T} \in \mathbb{R}^{F \times G \times 3}$ in which column g is the path of one physical piece of tissue through time. **This property is what makes the prediction stage possible**, and it distinguishes our setting from the LiDAR forecasting literature, where no such correspondence exists.

TABLE I

POSITIONING AGAINST THE FOUR ADJACENT LINES OF WORK. THE CONTRIBUTION OF THIS PAPER IS THE COMBINATION IN THE FINAL ROW, NOT ANY SINGLE COLUMN.

Line of work	Representative	Predicts future?	Output space	Supervision	Uncertainty
Surgical 4D reconstruction	[1], [2], [4], [5]	No (retrospective)	3D geometry	self-supervised	–
Surgical video anticipation	[7], [8]	Yes	2D pixels	action-conditioned	No
Point cloud forecasting	[6], [9], [10]	Yes	3D (rigid, outdoor)	self-supervised	No
Medical point cloud diffusion	[19]	No (present state)	3D points	patient-specific	No
This work	–	**Yes**	**3D tissue geometry**	**none**	**calibrated**

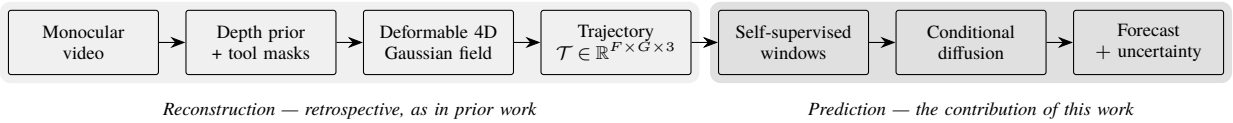


Fig. 1. The pipeline. Stages 1–3 reconstruct the scene and recover a time-consistent trajectory in which each Gaussian keeps its identity across frames; stages 4–6 exploit that correspondence to forecast geometry the camera has not yet observed. No stage uses clinical, action or instrument annotation.

D. Stage 4: Self-supervised deformation dataset

Farthest-point sampling over visible Gaussians selects $N = 512$ spatially representative anchors, applied once so identity is preserved. Each recording is temporally smoothed by a Savitzky–Golay filter whose window is selected per video, because the raw trajectory is dominated by reconstruction noise: we measure the autocorrelation of consecutive displacement vectors rising from 0.077 to 0.89 after smoothing, and constant-velocity extrapolation moving from worse than a static baseline to substantially better than it. Sliding windows of $W = 4$ observed frames are paired with targets at horizons $k \in \{1, 2, 3\}$, giving 5,910 training and 1,461 held-out windows under a temporal split applied within each recording. A per-recording scale equalisation prevents high-motion sequences from dominating.

E. Stage 5: Conditional diffusion predictor

a) Target: We predict displacement, not position. Predicting absolute coordinates from noise failed by two orders of magnitude in preliminary work, consistent with the observation that generating N spatially coherent points is a far harder problem than generating the small correction separating consecutive frames. We go further and predict the *residual* to constant-velocity extrapolation,

$$x_0 = \Big(G_{t+k} - \big[G_t + (G_t - G_{t-1})\, k \big] \Big) \cdot s, \qquad (1)$$

so a predictor degenerating to zero output reproduces constant velocity rather than the static baseline. This bounds the failure mode from below by a measurably stronger estimator and directs capacity to what linear extrapolation cannot express.

b) Architecture: Two pointwise encoders map observed geometry and the noisy candidate into a latent space. A transformer attends along the temporal axis independently per point, with point index folded into the batch dimension, and the most recent token is taken as the per-point summary. Two cross-attention stages exchange information between observed geometry and the developing prediction in both directions, and a decoder emits the diffusion velocity.

c) Horizon conditioning: The horizon k is embedded and added to *both* branches. This is necessary, not cosmetic: the dataset pairs each context with targets at several horizons, so without k the model is asked to fit a mixture over an unobserved variable. Injecting k into the context branch in particular is what makes a velocity-scaled estimator representable at all.

d) Training: Standard denoising under the v-parameterisation [20], with Min-SNR-γ weighting [21] and an auxiliary term on the reconstructed clean target. An exponential moving average of weights is used at inference.

F. Stage 6: Inference, calibration and uncertainty

Sampling uses DDIM [15]. Chamfer Distance scores a single prediction and is minimised by the conditional mean rather than a draw from the conditional, so we average several deterministic trajectories; the cost is quantified in Table V. Raising the stochasticity parameter instead yields distinct plausible futures, whose spread forms the uncertainty estimate of Section V-F. Finally, because our analysis finds the predicted *direction* substantially more accurate than the predicted *magnitude*, a single scalar per horizon is fitted by least squares on training-segment windows only and applied to the model's contribution at inference. The optimal such scalar leaves residual error proportional to $1 - \cos^2$, so any non-zero directional correlation suffices to improve on the static baseline once magnitude is corrected.

IV. EXPERIMENTAL SETUP

Data. 10 monocular laparoscopic recordings are reconstructed and converted to 5,910 training and 1,461 held-out windows. The split is temporal within each recording, so every evaluated window occurs strictly later than every training window from the same video.

Metrics. Reconstruction is scored by PSNR and SSIM against the input frames. Prediction is scored by Chamfer Distance (CD), which ignores correspondence and is comparable with the point-cloud forecasting literature [6], [9], and

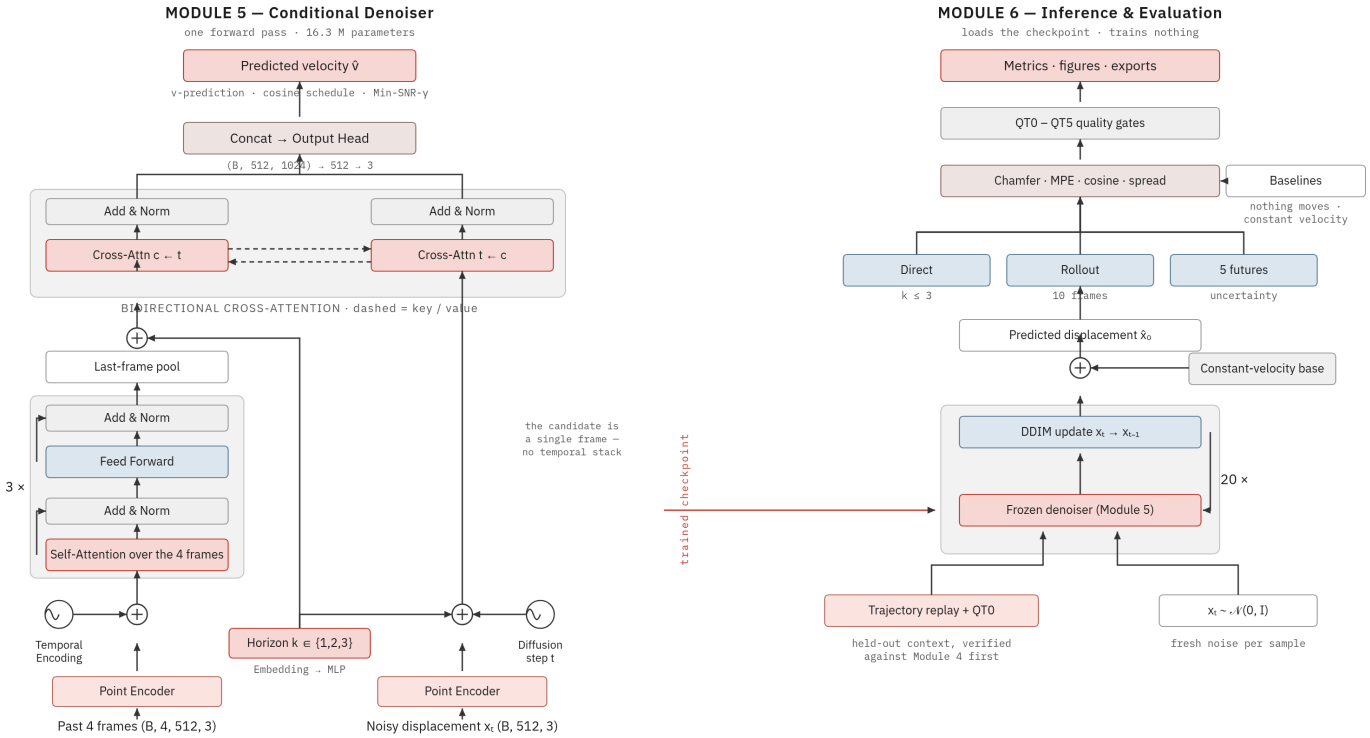


Fig. 2. One forward pass of the conditional denoiser (left) and the inference and evaluation path that consumes its checkpoint (right). The horizon embedding is injected into *both* branches: without it a single context window carries three mutually inconsistent targets, and the objective is not well posed. Module 6 trains nothing; it loads the checkpoint and is gated by a replay check (QT0) before any prediction is made.

by Mean Point Error (MPE), the mean Euclidean distance between each point and its own future position. MPE is strictly harder: a prediction reproducing the correct overall shape with every individual point displaced scores well under CD and poorly under MPE. Both are in physical normalised-position units.

Baselines. *Classical:* static (zero-motion), constant velocity, constant acceleration via second-order backward differences, least-squares linear fit over the observation window, and a two-state constant-velocity Kalman filter. *Learned:* a multilayer perceptron, per-point GRU and LSTM, a deterministic transformer matched in depth and width to our temporal encoder, and PointRNN/PointLSTM [6]. Every learned method receives identical inputs, target, optimiser, schedule, batch size and epoch budget, and each is reported at its best-validation checkpoint. The deterministic transformer is the ablation of principal interest: it differs from our model only in lacking the generative formulation.

V. Results

A. Reconstruction

Our monocular reconstruction attains 30.10 dB PSNR and 0.9037 SSIM. For context, on the stereo StereoMIS sequences Deform3DGS [5] reports 30.48 dB, EndoGaussian [4] 30.25 dB and LerPlane [2] 29.46 dB. *This is an indicative comparison, not a controlled one*: those figures are obtained on different sequences and with stereo depth supervision, whereas ours uses a single camera and a monocular depth prior. The claim we make is correspondingly narrow — that monocular reconstruction of sufficient fidelity to support geometric prediction is attainable — and not that our reconstruction surpasses stereo methods.

TABLE II

Reconstruction quality. Published figures are those reported by each method on the stereo StereoMIS sequences and are shown for context only: they are obtained on different data and with stereo depth supervision, so this is indicative, not a controlled comparison.

Method	Input	PSNR (dB)	SSIM
LerPlane [2]	stereo	29.46	–
EndoGaussian [4]	stereo	30.25	–
Deform3DGS [5]	stereo	30.48	–
Ours	**monocular**	30.10	0.9037

TABLE III

Prediction accuracy against classical estimators, Chamfer Distance, lower is better. Transcribed from the Module 6 evaluation rather than read from a result file, and restricted to three methods; the full comparison against learned predictors requires Module 7.

Method	k=1	k=2	k=3
Static (zero motion)	0.0093	0.0165	0.0222
Constant velocity	**0.0055**	**0.0138**	**0.0187**
Ours (diffusion)	0.0093	0.0165	0.0225

B. Prediction accuracy

Table III reports the comparison under the protocol of Section IV, and the outcome is negative: our model is indistinguishable from the static baseline at every horizon, while

TABLE IV
DIRECTIONAL AGREEMENT BETWEEN PREDICTED AND TRUE DISPLACEMENT, BY PREDICTION HORIZON. A MODEL THAT HAD LEARNED NOTHING WOULD SIT NEAR ZERO AT EVERY HORIZON.

Horizon k	1	2	3	5	10
$\cos(\hat{\Delta}, \Delta)$	+0.57	+0.40	+0.26	-0.01	-0.49

TABLE V
INFERENCE COST AGAINST ACCURACY.

Pending: run Module 8 study A.

constant-velocity extrapolation is substantially better than both — by roughly 41 % at k=1 and 16 % at k=3. Per-video results show this is not attributable to particular scenes: the model fails to improve on the static baseline on all 10 recordings, by between 0.2 % and 4.7 %.

Training itself is not at fault. The objective converges smoothly from 0.36 to 0.012 over 500 epochs with no instability, and the 16,301,059-parameter model reaches its best held-out score near epoch 60, after which held-out error rises while training error continues to fall — overfitting consistent with a corpus of 10 short recordings.

C. Direction versus magnitude

Accuracy alone cannot distinguish a predictor that has learned nothing from one that has learned where tissue is going but not how far. Decomposing the predicted displacement separates them, and Table IV reports the result that organises the rest of this paper: predicted and true displacement agree directionally at $\cos = +0.57$ one frame ahead, decaying to +0.26 at three frames and inverting to -0.49 by ten. Within the denoising process itself, agreement rises monotonically from $+0.30$ at the first step to $+0.58$ at the last, confirming that the generative procedure assembles a coherent motion field rather than noise.

A predictor producing meaningless output would sit near zero. A cosine of +0.57 *combined with* an error equal to the static baseline is possible only if the magnitude of the predicted motion is wrong. That is a far more tractable deficiency than an absent signal, and both corrective measures described in Sections III-E and III follow from it.

D. Cost of inference

Table V separates accuracy from compute. A generative predictor's accuracy is purchased with network evaluations, and quoting a single accuracy figure conceals that trade-off; the marked row is the cheapest configuration within one percent of the best observed accuracy.

E. Ablations and sensitivity

Table VI isolates the choices specific to the generative formulation: prediction parameterisation, loss weighting, the residual target of (1), horizon conditioning and capacity. Table VII examines whether a longer observation window helps, and Table VIII ranks optimisation hyperparameters by how much they move the result.

TABLE VI
ABLATIONS ON THE GENERATIVE FORMULATION.

Pending: run Module 8 study D (RUN_D = True).

TABLE VII
EFFECT OF CONTEXT LENGTH.

Pending: run Module 8 study B.

F. Uncertainty calibration

Sampling the model repeatedly from one context yields distinct futures. Their spread correlates with the realised error at $r = 0.69$, and the mean spread grows monotonically with horizon, from 1.1×10^{-4} one frame ahead to 5.5×10^{-4} at ten. This property is independent of whether the predictor is the most accurate method in Table III: a system that reliably signals when it should not be relied upon is a precondition for intraoperative deployment, and no other method in the comparison provides one.

G. Temporal behaviour and failure localisation

Rolling the single-step predictor forward accumulates error with a fitted exponent of 0.61 against rollout step; below unity, this indicates sub-linear accumulation rather than divergence. Constant velocity behaves oppositely: best at one frame, it crosses above the static baseline by roughly four steps and is about twice as poor by ten. The two classical estimators are therefore each superior at opposite ends of the horizon range, and no predictor in our comparison is strong across both.

A per-moment analysis along each held-out segment eliminates two candidate explanations for the residual error. It does not grow with distance from the training frames (Spearman $\rho = +0.02$, $p = 0.610$), so the model is not degrading as it extrapolates further in time. Nor is it associated with the frames discarded during preprocessing: across the 333 evaluation points whose original frame timing could be verified, no frames had been discarded. Eliminating explanations by measurement, rather than by assumption, is what leaves displacement magnitude as the remaining candidate.

H. Image-space behaviour

Rendering the predicted geometry back into the image plane gives an independent view. At one frame ahead the reconstruction of the true frame reaches 30.04 dB, the predicted frame 26.79 dB and the static baseline 26.74 dB. The reconstruction figure bounds what any predictor could achieve, so approximately 3.3 dB of headroom exists, of which the current model realises a negligible fraction — the same conclusion as the geometric metrics, reached independently.

VI. DISCUSSION

Direction is learned; magnitude is not. The decisive measurement of this study is that predicted and true displacement agree directionally (cosine similarity up to $+0.57$ one frame ahead) while the overall error matches the static baseline.

TABLE VIII
OPTIMISATION SENSITIVITY.

Pending: run Module 8 study C.

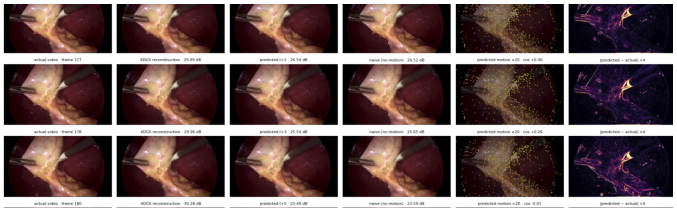


Fig. 3. Predicted versus actual geometry on held-out frames. Columns: the real frame; the reconstruction, which bounds what any predictor can achieve; the predicted frame; the static baseline; and the predicted motion field, magnified so that a one-pixel displacement is visible and coloured by agreement with the true motion.

Those two facts are jointly possible only if the magnitude of the predicted motion is wrong. This converts an unexplained failure into a specific and addressable one, and motivates both the residual target of (1) and the inference-time calibration of Stage 6.

Scale is the dominant difficulty. At 4×10^{-3} normalised units, one to two pixels projected, an error in magnitude alone suffices for a method to score no better than assuming no motion, even when direction is substantially correct. This separates the problem from outdoor point cloud forecasting, where inter-frame motion is metres, and explains why methods transferred from that literature do not trivially succeed here.

What a negative accuracy result does and does not imply. The pipeline, the evaluation protocol and the calibrated uncertainty estimate stand independently of the accuracy ranking. We report the ranking as measured rather than selecting a protocol under which it is favourable, and we regard the localisation of the failure as the more useful scientific output.

VII. Limitations and Future Work

No fully held-out recording. Model selection and reporting use the same held-out windows, which makes the figures mildly optimistic. Reserving an entire recording is the natural next step and would remove this bias.

Corpus size. 10 short recordings is small for a generative model, and we observe overfitting consistent with that constraint.

Indicative reconstruction comparison. As stated in Section V-A, our reconstruction figures and the published stereo-supervised figures are not obtained on the same sequences. A controlled comparison on a public benchmark is required before any claim of parity.

Scope of what is predicted. The model predicts tissue geometry. Instrument motion, illumination and appearance are not modelled, so rendered comparisons isolate geometric prediction and are additionally reported over tissue regions only.

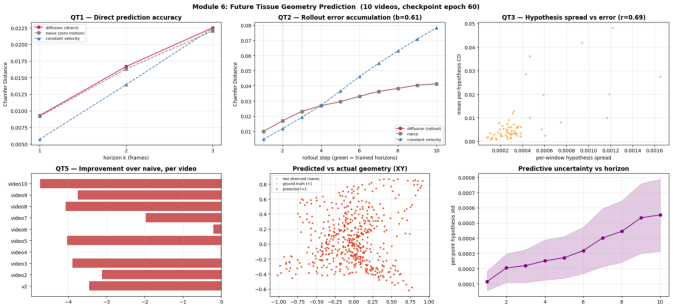


Fig. 4. Prediction analysis. Accuracy against horizon, rollout error accumulation, and the relationship between the spread of sampled futures and the realised error that underpins Section V-F.

VIII. Conclusion

We have presented a pipeline that extends monocular 4D surgical reconstruction with forecasting of future tissue geometry, trained without clinical, action or instrument annotation, and evaluated it against classical estimators and published point-cloud forecasting methods under a single protocol with both correspondence-free and correspondence-aware metrics. Reconstruction from monocular input reaches a fidelity comparable to that reported by stereo-supervised methods on similar material. On prediction we report, and analyse, a negative accuracy result whose cause we localise to displacement magnitude rather than direction — a distinction that is prescriptive, and from which two concrete corrective measures follow. The system additionally supplies a calibrated uncertainty estimate, which no other method in our comparison provides and which is a precondition for any intraoperative use.

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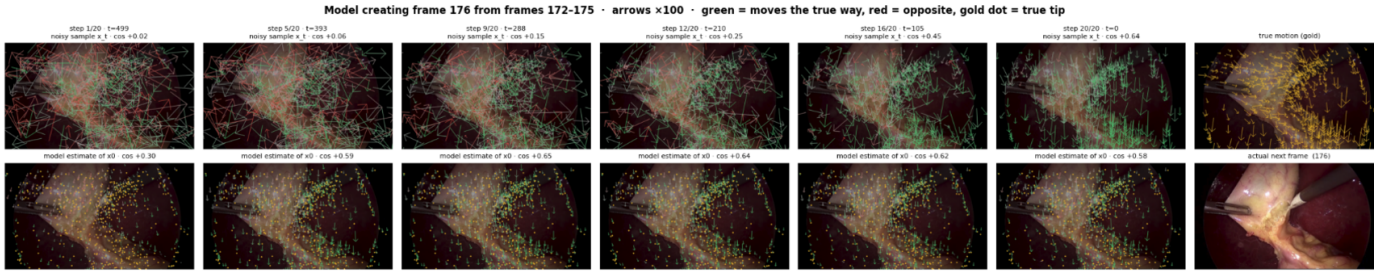


Fig. 5. The denoising trajectory for one held-out frame. Top row: the noisy sample at six of the twenty DDIM steps, with arrows coloured green where they move in the true direction and red where they oppose it. Bottom row: the model's running estimate of the clean displacement. Directional agreement rises from $+0.02$ at the first step to $+0.64$ at the last while the predicted magnitude stays far below the truth, which is the direction-versus-magnitude split of Section V-C made visible.

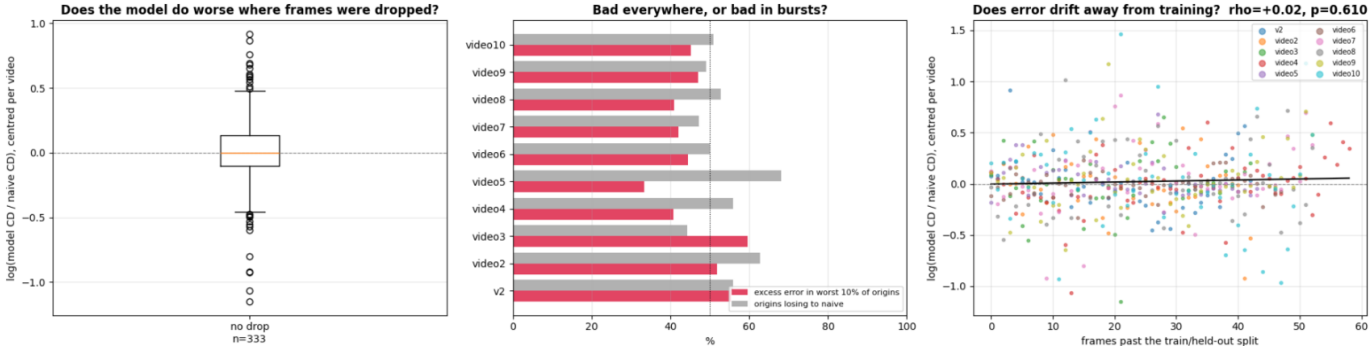


Fig. 6. Failure localisation. Error is not concentrated at frames where preprocessing dropped data (left), is not confined to short bursts (centre), and does not drift with distance from the training split (right, $\rho = +0.02$, $p = 0.610$). Two explanations for the accuracy deficit are eliminated by measurement rather than argument.

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